USN. | | | | | | | | | | [24MTSCSEDE14]

# FIRST SEMESTER M.Tech EXAMINATION, JANUARY-2025

## APPLIED-AI

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:**
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

| | | | CO's | Blooms Level | Marks |
|---|---|---|---|---|---|
| **1.** | **a.** | In a warehouse robot navigation system, the robot needs to find the shortest path to pick items from shelves. Compare and contrast the suitability of DFS, BFS, and A* algorithms for this task, considering constraints like time and memory. | **CO1** | **4** | **10** |
| | **b.** | Design a knowledge representation model to handle customer support queries in a retail application. Propose how reasoning mechanisms could be integrated to provide automated responses. | **CO1** | **3** | **10** |
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| **2.** | **a.** | A retail company wants to predict future sales based on past data. Develop a supervised learning model for this task, detailing the steps for feature selection, training, and evaluation. | **CO2** | **3** | **10** |
| | **b.** | Build an automated system to detect fraudulent transactions in a banking system. Design a deep learning architecture for this use case, including the data preprocessing and performance metrics you would use. | **CO2** | **3** | **10** |
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| **3.** | **a.** | A company wants to implement sentiment analysis for movie reviews in multiple languages. Explain the challenges they might face in preprocessing and word embeddings and suggest methods to overcome them. | **CO3** | **3** | **10** |
| | **b.** | Design a chatbot for a hospital to help patients book appointments and answer FAQs. Outline the architecture, including preprocessing, model training, and how the bot will handle ambiguous queries. | **CO3** | **3** | **10** |
| | | **OR** | | | |
| **4.** | **a.** | A company is analyzing customer feedback using NLP. Define tokenization and explain its importance in breaking down the sentence *"The product quality is excellent but delivery was delayed"*. How many tokens will be generated using space as a delimiter? | **CO3** | **2** | **10** |
| | **b.** | A social media platform uses sentiment analysis for targeted | **CO3** | **3** | **10** |

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| | | advertising. Evaluate the impact of misclassifying a neutral review, such as *"The service is okay"*, as negative or positive. Suggest a mitigation strategy based on model confidence scores. | | | |
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| **5.** | **a.** | Develop an image segmentation system for detecting tumors in MRI scans. Explain the choice of algorithm, dataset preparation, and how you would evaluate its accuracy and reliability. | **CO4** | **4** | **10** |
| | **b.** | Design a real-time object detection system for autonomous drones to identify obstacles in outdoor environments. Detail the architecture and how you would ensure the model performs effectively in varying lighting and weather conditions. | **CO4** | **4** | **10** |
| | | **OR** | | | |
| **6.** | **a.** | A security system uses CNNs for facial recognition. Describe the function of convolutional layers in extracting features such as edges and textures from an image. Suppose the system processes 1000 images in 5 minutes; calculate the average number of images processed per second. | **CO4** | **3** | **10** |
| | **b.** | A retail store wants to implement object detection for counting products on shelves. Explain the difference between object detection and image segmentation. If there are 10 objects in an image and the detection system has an accuracy of 80%, how many objects are likely to be correctly detected? | **CO4** | **3** | **10** |
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| **7.** | **a.** | A manufacturing company is using IoT sensors to monitor equipment. Propose an AI-based predictive maintenance system, highlighting the use of sensor data, AI models, and deployment strategies. | **CO5** | **4** | **10** |
| | **b.** | A financial institution wants to detect money laundering activities using AI. Design a solution that integrates anomaly detection algorithms and real-time monitoring to identify suspicious transactions. | **CO5** | **4** | **10** |
| | | **OR** | | | |
| **8.** | **a.** | A hospital uses AI for predicting patient readmission. Define predictive analytics and identify which features (e.g., age, diagnosis, previous visits) are crucial. If a logistic regression model predicts readmission with 85% accuracy, how many correct predictions are expected out of 200 patients? | **CO5** | **3** | **10** |
| | **b.** | A smart factory uses IoT sensors to monitor machines and reduce downtime. Describe how AI processes real-time sensor data to predict failures. If a machine breaks down 10 times annually and predictive maintenance reduces breakdowns by 40%, how many breakdowns can be prevented? | **CO5** | **3** | **10** |